

UNIVERSITEIT UTRECHT

MASTER THESIS

Jump-Starting Evidence Synthesis

Initializing Active Learning Models for Systematic
Reviews using LLM-generated Data

Author:

Timo van Ommeren

Supervisors:

Prof. dr. A.G.J. (Rens) van de Schoot, Lauke Stoel

MSc Methodology and Statistics for the Behavioural,
Biomedical and Social Sciences

Methodology Department, Utrecht University



February 3, 2026
Word count: 4395

It is better to take many small steps in the right direction than to make a great leap forward only to stumble backward. - Chinese Proverb

1 Introduction

1.1 Background: AI-assisted systematic reviews

Researchers and practitioners are continually challenged to base their decisions on the latest scientific evidence. However, the ever-increasing volume of scientific literature makes it difficult to keep up with the latest findings (Bornmann and Mutz 2015; Jinha 2010; Bastian, Glasziou, and Chalmers 2010). To address this challenge, evidence synthesis methods such as systematic reviews and meta-analyses have been developed to help summarize and interpret large bodies of scientific literature (Grant and A. Booth 2009; Chalmers, Hedges, and Cooper 2002).

Systematic reviews aim to synthesize all relevant evidence on a specific research question in a transparent and reproducible manner by systematically searching, screening, and critically appraising the literature (Higgins, Green, et al. 2008; Brignardello-Petersen, Santesso, and Guyatt 2025). A meta-analysis is a statistical technique often used within systematic reviews to quantitatively combine the results of multiple studies (Borenstein et al. 2021). Both techniques contribute to evidence-based practice by providing comprehensive and unbiased summaries of the available evidence on a given topic.

A significant constraint to the widespread adoption of systematic reviews is the considerable time and resources required to conduct them (Nussbaumer-Streit et al. 2021). Specifically, the screening of titles and abstracts to identify relevant papers is a major bottleneck.

Fortunately, recent advances in machine learning have produced tools that assist researchers in screening papers by prioritizing the most relevant papers for screening first. One such tool is *ASReview*, an open-source software package for AI-assisted systematic reviews (Van De Schoot et al. 2021). *ASReview* uses active learning models to iteratively learn from the user’s screening decisions and adaptively prioritize the most relevant papers for screening. Moreover, recent workflows have integrated *ASReview* with large language models (LLMs) as validating second screeners to further speed up the screening process and reduce human error (J. d. Bruin et al. 2025; Bron et al. 2024).

AI-assisted screening can significantly reduce the time and effort needed to conduct systematic reviews and may even help identify more relevant papers compared to traditional screening methods (Teijema et al. 2025; Byrne et al. 2024; Ferdinands et al. 2023; Schoot et al. 2025). Thus, AI-assisted screening can enhance the efficiency and accessibility of systematic reviews, thereby improving evidence-based decision-making in various fields (Bos 2025; Ge et al. 2024; Harrington 2017; Powers and Carlbring 2016).

1.2 Research gap: the cold start problem

A key limitation to the efficiency and accessibility of AI-assisted screening tools, and consequently their contribution to evidence-based decision-making, is *the cold start problem*.

The cold start problem refers to the difficulty that certain recommendation algorithms have in providing accurate recommendations when dealing with new users or items (Panda and S. Ray 2022; Lika, Kolomvatsos, and Hadjiefthymiades 2014). In the context of AI-assisted systematic reviews, the cold start problem arises when the active learning model has no prior examples of relevant papers to learn from at the start of the screening process (Van De Schoot et al. 2021). As a result, the active learning model may struggle to accurately prioritize relevant papers for screening, leading to suboptimal starting performance

and increased screening time.

In general, most recommender systems circumvent a cold start by using multiple complementary recommendation algorithms, leveraging external data sources, or querying users for additional information (Bobadilla et al. 2013; Su and Khoshgoftaar 2009). For example, when signing up for Instagram, a user may be shown a list of celebrities and influencers to follow or be asked to add external data, such as their Facebook account or phone contacts.

Similarly, the cold start problem rarely arises in AI-assisted systematic reviews. Most systematic reviews are based on previous research. In clinical medicine, for example, similar studies using randomized controlled trials (RCTs) on specific interventions for specific conditions are constantly being published (Moosapour et al. 2021). It is therefore easy to find prior research on similar topics that can be used to initialize the active learning model and avoid a cold start.

Moreover, replications of systematic reviews have become so common in some fields that there are now systematic reviews of systematic reviews and meta-meta-analyses. Some specialized research teams have opted to maintain continuously updated *living systematic reviews* (Riaz et al. 2024; Cuijpers et al. 2023). In these cases, the cold start problem has dissappeared entirely.

The cold start problem does pose an issue in AI-assisted systematic reviews when the evidence base is less established or rapidly evolving. In these cases, there may be no obvious precedents for the review at hand.

Indeed, fields with less established or rapidly evolving evidence bases pose a challenge to systematic reviews more generally. These fields often consist of heterogeneous studies with varying methodologies, populations, and outcomes, making it difficult to synthesize findings in a meaningful way. Consequently, researchers may opt to conduct an initial mapping or scoping review before undertaking a full systematic review (Grant and A. Booth 2009; Abou-Setta et al. 2016). Alternatively, researchers may forego a systematic literature review altogether and rely on expert opinion, narrative reviews, or case-based reasoning for decision-making (Aamodt and Plaza 1994; Evans 2022).

For example, at the start of the COVID-19 pandemic, there was limited knowledge about the the SARS-CoV-2 virus. Practitioners and policymakers had to make decisions based on limited, rapidly changing evidence, relying on expert opinion, preliminary studies, and other gray literature, rather than systematic reviews (Kurzer and Ornston 2023; Evans 2022).

Rapid reviews have been developed as an alternative to systematic reviews to improve evidence-based decision-making in such situations (Tricco, Langlois, Straus, et al. 2017; Tricco, Garritty, et al. 2020). Rapid reviews aim to provide timely evidence synthesis by streamlining or omitting certain steps of the systematic review process (Smela et al. 2023; Grant and A. Booth 2009). However, rapid reviews necessarily sacrifice methodological rigor for speed, which increases the possibility of human error and bias (Haby et al. 2016).

Another example of fields where the use of systematic reviews is challenging is in rapidly evolving fields such as artificial intelligence (AI), new psychoactive substances (NPS) (Evans-Brown and Sedefov 2018), or emerging infectious diseases (Mehand et al. 2018). In these fields the risk that findings become outdated by publication significantly limits the appeal of conducting systematic reviews (Mossel et al. 2025).

Thus, there is a clear need to improve the efficiency and accessibility of systematic reviews in fields where the evidence base is less established or rapidly evolving. This may in part be achieved by the application of AI-assisted screening tools in these fields

(Gartlehner, Nussbaumer-Streit, et al. 2025). However, a key limitation to the application of AI-assisted systematic reviews in fields with less established or rapidly evolving evidence bases is the need for prior examples of relevant papers to initialize the active learning model and avoid a cold start.

In summary, we may conclude that at the core of science is an evidence base that is continually being established and evolved. These fields have well established methodologies and systematic reviews have become common place, or at the very least straight forward to conduct. Accordingly, AI-assisted systematic reviews do not face a cold start problem in these fields and can be implemented with relative ease.

In contrast, at what we may call the edges of science or the scientific frontier, the evidence base is less established or rapidly evolving (Funtowicz and Ravetz 2018). In this case, systematic reviews are more challenging to conduct, and, consequently, less common and less methodologically rigorous. Accordingly, AI-assisted systematic reviews may face a cold start problem further limiting their appeal and contribution to evidence-based decision making in these fields.

To conclude, the cold start problem is a key limitation to more widespread adoption of AI-assisted systematic reviews and evidence-based decision making.

1.3 Proposed approach: leveraging LLMs

1.3.1 A brief overview of large language models

Large language models (LLMs) are a type of generative AI based on transformer architectures that use self-attention mechanisms to capture long-range dependencies in text data (Vaswani et al. 2017; Devlin et al. 2019; Radford et al. 2019). In contrast to most classical machine learning techniques, LLMs are therefore well suited to handle the complexities and nuances of human language (Brown et al. 2020; Bommasani 2021).

Furthermore, most LLMs on offer have been pre-trained on virtually all of human knowledge available digitally, ranging from academic books to Reddit discussions (Achiam et al. 2023; Touvron et al. 2023). On top of this, these LLMs have been tailored through reinforcement learning from human feedback to align their outputs with human preferences (Ouyang et al. 2022; Perrigo 2023). In this sense, interacting with an LLM is analogous to seamlessly communicating with a human-tailored version of the entirety of human knowledge (Farrell et al. 2025).

Unsurprisingly, the 'chat' version of OpenAI's LLM, ChatGPT, became the fastest growing consumer application in history, reaching over 100 million users within two months of its release (Hu 2023). Similarly, LLMs have been rapidly adopted in various professional fields, including healthcare (Singhal et al. 2023), law (Katz et al. 2024), and education (P. Zhang and Tur 2024). The advent of LLMs has even renewed discussions about the future of work (Brynjolfsson, D. Li, and Raymond 2025), and the possibility of imminent superintelligent AI (Nick 2014; Agrawal, Gans, and Goldfarb 2024).

However, the hype surrounding LLMs has also been met with significant pushback. A thorough examination of the history of AI reveals that similar waves of enthusiasm have occurred in the past, only to be followed by periods of disillusionment and reduced funding, known as 'AI winters' (Tan 2017; Crevier 1993; Mitchell 2021). Critics argue that much of the current excitement around LLMs is based on the misconception that these models possess genuine understanding and reasoning capabilities, when in reality

they are primarily sophisticated pattern-matching algorithms (Bender et al. 2021; Marcus and Davis 2019; Mitchell 2024; Wei et al. 2022; Stevenson et al. 2024).

Indeed, a recent MIT report found that 95% of generative AI projects fail to deliver bottom-line results (NANDA 2025). The report concluded that successful projects distinguished themselves by having clear problem definitions and objectives for the use of generative AI, rather than adopting generative AI for its own sake.

The application of LLMs thus merits careful consideration. Not in the least because LLMs pose significant challenges to the scientific method itself (Hosseini and Horbach 2023; Thorp 2023; Noy and W. Zhang 2023). Most LLMs are owned by proprietary companies (Ollion et al. 2024) and do not adhere to human-in-the-loop or explainable AI principles (Mosqueira-Rey et al. 2023; Arrieta et al. 2020). Furthermore, due to the inherent stochasticity of LLMs, workflows that rely on LLMs complicate reproducibility (Steyvers and Kumar 2024; Shyr et al. 2025).

Finally, LLMs also raise significant ethical concerns. LLMs have been found to reproduce and even amplify societal biases present in their training data (E. Sheng et al. 2019; Bolukbasi et al. 2016; Caliskan, Bryson, and Narayanan 2017). Moreover, the environmental impact of training and deploying large-scale LLMs is substantial, raising questions about the sustainability of their widespread use (Strubell, Ganesh, and McCallum 2019; Jeanquartier et al. 2026; S. Ren et al. 2024).

All in all, LLMs represent a powerful yet controversial technology that has the potential to transform various aspects of society. However, their adoption must be approached with caution, considering both their capabilities and limitations.

1.3.2 The application of LLMs to systematic reviews

In light of the rapid adoption of LLMs in various fields, it is perhaps unsurprising that they have already been applied to address challenges in all stages of the systematic review process (Mossel et al. 2025; Gartlehner, Nussbaumer-Streit, et al. 2025; Scherbakov et al. 2025).

The applications of LLMs range from search query generation (Shyr et al. 2025) to the screening of papers (Kim et al. 2025; Bron et al. 2024; Delgado-Chaves et al. 2025; Thode, Iftikhar, and Mendez 2025) to data extraction (Khan et al. 2025; Sun et al. 2024). Consequently, it is now in principle possible to use LLMs to conduct entire systematic reviews with minimal human involvement (X. Chen and Xue Zhang 2025; Cao et al. 2025; Brett and Myatt 2025; Han 2025).

Notably, however, there seems to be a lack of systematic assessment of the performance of LLMs in systematic reviews. Many studies appear to be either proof-of-concept studies or focus primarily on the technical feasibility. Accordingly, one scoping review found that most studies applying LLMs to systematic reviews were preliminary and lacked rigorous evaluation (Lieberum et al. 2025). The conclusion seems to be that while LLMs can significantly reduce the workload of systematic reviews, the use of LLMs may compromise the quality and trustworthiness of systematic reviews (Sollini et al. 2025; Cao et al. 2025).

That said, one clear pattern does seem to emerge from the existing literature: LLM-human collaboration appears to outperform both LLMs alone and humans alone in various stages of the systematic review process (Haichao Chen et al. 2025). For example, using LLMs as validating second screeners seems to improve screening performance compared to either LLMs alone or humans alone (Helms Andersen et al. 2025; Schoot et al. 2025). This similarly holds true for data extraction (Lai et al. 2025). Tellingly, this finding is in line with guidelines proposed for the responsible use of AI in systematic reviews, which

emphasize the need for human oversight and transparency (Page et al. 2021; Flemyng et al. 2025).

All in all, it’s clear that the field of systematic reviews is changing rapidly, and the application of AI in general, as well as LLMs in particular, shows great promise. However, the right balance between automation and human involvement has yet to be found, and more critical assessments are needed before LLMs can be widely adopted for systematic reviews (O’Connor et al. 2024; Gartlehner, Kahwati, et al. 2025).

It is worth noting that while LLMs aren’t perfect, neither are humans. Advocates of self-driving cars tend to claim that one day letting error-prone humans drive cars will be unimaginable (Parasuraman and Riley 1997; Tran 2016; Shariff, Bonnefon, and Rahwan 2017). This claim may be speculative but, in fact, in commercial aviation, autopilot systems with human oversight have already become the norm (Gouraud, Delorme, and Berberian 2017). Notably, this has made flying significantly safer (Eurostat 2026; Europe 2017). It should therefore come as no surprise if largely automated systematic reviews with human oversight similarly become the norm in the near future.

In summary, while LLMs have already been applied to various stages of the systematic review process, their use must be approached with caution to ensure that systematic reviews remain transparent, reproducible, and trustworthy.

1.3.3 How LLMs have been applied to active learning and recommender systems.

The cold start problem is a promising application area for LLMs. In recommender systems, LLMs have been used to generate synthetic user profiles or item descriptions to overcome the cold start problem (Xiangyu Zhang et al. 2023). By leveraging the vast knowledge encoded in LLMs, these synthetic profiles can provide valuable information to initialize recommendation algorithms, thereby improving their performance in cold start scenarios.

Y. Xia et al. (2025) provide a comprehensive overview of how LLMs can be used in active learning.

(Bayer and Reuter 2024) propose ActiveLLM, a framework that integrates LLMs into active learning workflows. ActiveLLM uses LLMs to generate informative queries and select the most uncertain samples for labeling, thereby improving the efficiency of the active learning process. The authors demonstrate that ActiveLLM outperforms traditional active learning methods on several benchmark datasets.

Exploiting epistemic uncertainty in recommender systems (Xiang et al. 2025)

Maximum entropy approach within ASReview.

1.4 Objectives

The need for prior examples of relevant papers to initialize the active learning model and avoid a cold start is a key limitation to more widespread adoption of AI-assisted systematic reviews and evidence-based decision making. Furthermore, LLMs have shown promise in addressing the cold start problem in recommender systems and active learning. The main objective of this study is therefore to investigate whether LLMs can be used to overcome the cold start problem in AI-assisted systematic reviews.

The main objective of this study will be investigated through a simulation study using ASReview. The need for prior examples of relevant papers to initialize the active learning model will be addressed by generating synthetic abstracts using an LLM based on the

inclusion and exclusion criteria of the systematic review at hand. The performance of ASReview initialized with LLM-generated abstracts will be evaluated based on its starting performance, defined as the number of relevant papers successfully retrieved from among the first 100 screened papers. The starting performance in the LLM condition will be compared to that of three control conditions: the cold-start condition (no prior relevant abstracts), the true-example condition (one relevant abstract sampled from the actual relevant abstracts), and the criteria-only condition (the inclusion and exclusion criteria directly function as an example of a relevant abstract).

Formally, the main objective of this study can be defined as testing the following null hypotheses:

$$\begin{aligned} H_{0,\text{LLM-ic}} &: \mu_{\text{llm}} - \mu_{\text{ic}} = 0, \\ H_{0,\text{LLM-te}} &: \mu_{\text{llm}} - \mu_{\text{te}} = 0, \\ H_{0,\text{LLM-cs}} &: \mu_{\text{llm}} - \mu_{\text{cs}} = 0. \end{aligned}$$

In which the subscripts *llm*, *co*, *te*, *cs* stand for the conditions: LLM, criteria-only, true-example, and cold-start, respectively; and where μ_i is the true mean starting performance in condition *i*. The alternative hypotheses are all two-sided.

Finally, an exploratory analysis will be conducted to investigate the effect of several secondary (auxiliary?) independent variables on starting performance. These include: LLM-specific variables (e.g., number of abstracts generated), systematic review-specific variables (e.g., topic of the systematic review), inclusion/exclusion criteria-specific variables (e.g., presence of abbreviations). It will also be attempted to investigate the precision (or overdispersion) in starting performance across the four conditions.

2 Methodology

2.1 Simulation study design

The goal of the simulation study is to investigate whether LLMs can be used to overcome the cold start problem in AI-assisted systematic reviews. This will be done by simulating the abstract-title based screening of 23 previously screened and labelled systematic reviews from the SYNERGY collection using ASReview¹ (Van De Schoot et al. 2021; J. d. Bruin et al. 2025; De Bruin et al. 2023). The simulation study consists of one experimental condition and three control conditions (described below). The outcome variable is starting performance, defined as the number of relevant papers successfully retrieved from among the first 100 screened papers.

To start with the outcome variable, starting performance. Starting performance is defined as the number of relevant papers successfully retrieved from among the first 100 screened papers. This outcome variable was chosen based on the idea that approximately 100 papers can be screened in an hour (Nussbaumer-Streit et al. 2021) and as a practical way to limit time and computing power needed in each simulation run. It’s worth noting

¹Two sequential configurations of ASReview are used: the first queries papers at random until at least one relevant and one irrelevant paper have been found and ASReview’s classifier can be trained. Next the U4 configuration is used for the remainder of the screening process. Notably, however, in all conditions except for the cold-start condition, the first configuration is effectively skipped since at least one relevant paper is provided before screening begins and irrelevant papers are plentiful in all systematic reviews.

that simply counting the number of relevant papers found within the first 100 screened papers disregards the dynamic nature of the screening process². However, this simplifying assumption should suffice for the purpose of this study.

Considerable variation in starting performance is expected between the simulated systematic reviews. This variation is due to differences in topic, the percentage of relevant papers returned by the search, and abstract style of the systematic reviews (De Bruin et al. 2023). These differences lead to variations in the performance of ASReview across systematic reviews. Furthermore, some systematic reviews contain fewer than 100 relevant papers, which may be entirely included within the first 100 screened papers in certain simulation runs. This could result in ceiling effects in starting performance for these systematic reviews. See Appendix A for a list of all systematic reviews including their topic, total number of records, number of records included and the percentage of relevant records. Systematic review may therefore be considered a blocking factor or random effect in the simulation study design.

The main independent variables are the four conditions: one experimental condition and three control conditions. The experimental condition is from hereon referred to as the *LLM* condition. The three control conditions are the *cold-start* condition, the *true-example* condition, and the *criteria-only* condition.

The LLM condition represents the setting in which researchers aim to conduct a systematic review and have thoroughly defined the inclusion and exclusion criteria for their systematic review but lack examples of relevant papers to initialize the active learning model and avoid a cold start. The researchers in question solve their predicament by providing their inclusion and exclusion criteria to the LLM of their choosing with the request to generate the title and abstract of a hypothetically paper relevant to their systematic review.

This setting is simulated by taking the inclusion and exclusion criteria of a published systematic review and providing it to OpenAI’s GPT-4o-mini model with the request to generate at least one set of titles and abstracts of a paper relevant to the review at hand³. These sets of titles and abstracts are then given to ASReview along with the published systematic review to screen. See 1 for a schematic overview of the simulation pipeline.

The three control conditions serve to investigate (1) whether LLM-generated examples improve starting performance compared to a cold start, (2) whether LLM-generated examples perform comparably to actual examples of relevant papers, and (3) whether the LLM adds any value beyond the direct use of the inclusion and exclusion criteria.

Concretely, the three control conditions are operationalized as follows. In the cold-start condition, no papers are provided prior to the start of screening. In the true-example condition, one relevant paper is sampled from the set of actually relevant papers in the systematic review (with replacement between runs). In the inclusion-criteria condition, the inclusion and exclusion criteria of the systematic review function directly as an example of an abstract of a relevant paper with which to initialise the LLM, rather than using the LLM to generate an example. The three control conditions are compared to the LLM condition in a pairwise manner.

Within a subset of the conditions, several independent variables are manipulated to investigate their effect on starting performance.

First, three LLM-specific variables are manipulated in the LLM condition to investi-

²See discussion section ... for more details on this point, as well as directions for future research

³OpenAI’s gpt-4o-mini model is instructed via a DSPY module for dynamic prompt generation in each run (Khattab et al. 2023)

gate how the LLM’s configuration affects starting performance: (1) number of abstracts generated per simulation run (1, 4 or 7 abstracts), (2) length of abstracts generated per simulation run (100, 500, 900 words), and (3) the temperature (diversity) setting on the LLM (0.0, 0.4, 0.8). A full factorial design is used to investigate all combinations of these three variables (i.e., $3 \times 3 \times 3 = 27$ combinations) within the LLM condition (Burton et al. 2006; Morris, White, and Crowther 2019).

Second, within the LLM condition and the criteria-only condition, several aspects of the inclusion/exclusion criteria are manipulated. Specifically, the presence of exclusion criteria, expansion of abbreviations, and expansion of generic terms to include hyponyms and synonyms using MeSH are manipulated (For example, "rodents" may be expanded to include hyponyms such as "mice" and "rats", and "Wilson’s disease" to include synonyms such as "Progressive Lenticular Degeneration". See Appendix ?? for a complete overview of the manipulations). A multi-arm parallel group design is used to investigate the effect of these manipulations within the LLM condition and the inclusion/exclusion criteria condition (i.e., 4 parallel conditions) (Jaki and Vasileiou 2017).

Finally, there are several covariates that may influence starting performance. These include the characteristics of the systematic reviews (see Appendix A for a list of all systematic reviews including their characteristics) and aspects of the inclusion/exclusion criteria that are measured but not manipulated, including the length of the criteria (word count) and the complexity of the inclusion/exclusion criteria (e.g., use of logical operators such as AND, OR, NOT).

In summary, the simulation study consist of a total of four conditions (one experimental and three control conditions) with starting performance as the outcome variable. Within LLM condition the number of abstracts, the abstract length, and LLM’s temperature setting are manipulated using a full factorial design. Within the LLM condition and the inclusion/exclusion criteria condition, the presence of exclusion criteria, expansion of abbreviations, and expansion of generic terms are manipulated in a multi-arm parallel group design. Finally, several covariates are measured that may influence starting performance, including characteristics of the systematic review itself and characteristics of the inclusion/exclusion criteria. We can thus speak of a multi-arm parallel group design with a nested factorial and a nested multi-arm design in a subset of the arms.

The total number of runs was 49,680 (23 datasets $\times$ 4 conditions $\times$ 5 replications $\times$ 4 inclusion/exclusion criteria manipulations $\times$ 27 LLM manipulations). See appendix section B for a complete overview of the simulation runs and data formatting procedures.

2.2 Estimands

The primary estimands of interest are (1) the differences in mean starting performance between the LLM condition and each of the three control conditions, (2) the main effects and interactions between the three LLM-specific variables (number of abstracts, abstract length, and temperature) on starting performance within the LLM condition, and (3) the effect of the inclusion/exclusion criteria manipulations on starting performance within the LLM condition and the inclusion/exclusion criteria condition, as well as their interaction with condition (LLM vs. inclusion/exclusion criteria).

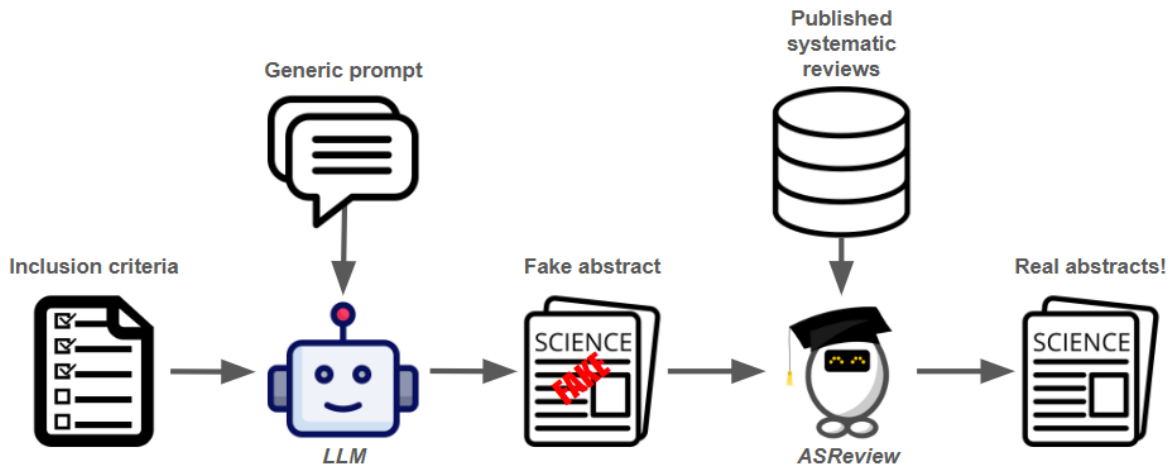


Figure 1: Simulation pipeline (for more detailed schematic of workings within ASReview see (J. d. Bruin et al. 2025))

2.3 Analysis

The purpose of the analysis is to investigate whether there are any statistically significant differences in starting performance between the LLM condition and each of the three control conditions, as well as to investigate the effect of several secondary independent variables on starting performance.

The outcome variable starting performance is defined as the number of relevant papers successfully retrieved from among the first 100 screened papers. This outcome variable may be treated as discrete proportion data, X/n , where $n \leq 100$. In this case a binomial model may be used. However, it is known that the binomial model may be approximated using the gaussian distribution when np and $n(1 - p)$ are both greater than 5 (Agresti 2002). This assumption is likely to hold when starting performance is averaged across multiple systematic reviews and replications, in which case the simpler linear model may be appropriate. However, if starting performance is modelled at the level of each systematic review and condition combination, a generalized linearmodel with a binomial family may be more appropriate.

The main marginal effect of condition on starting performance will be investigated extracted from a linear model using the *emmeans* package while controlling for multiple testing using Dunnett’s adjustment (Dunnett 1964; Sauder and DeMars 2019; Bretz, Hothorn, and Westfall 2016). A parametric bootstrap approach may be used as a confirmatory sensitivity check Sauder and DeMars 2019; Alver and G. Zhang 2024. Note that the LLM-specific variables are mean-centered to facilitate interpretability of the results, while the inclusion/exclusion criteria manipulations are set to their reference levels for the main effect of condition (i.e., presence of exclusion criteria = no, expansion of abbreviations = no, expansion of generic terms = no).

The main and interaction effects of the three LLM-specific variables (number of abstracts, abstract length, and temperature) on starting performance will be investigated by including these variables as covariates in the linear model described above. Similarly, the effect of the inclusion/exclusion criteria manipulations on starting performance within the LLM condition and the inclusion/exclusion criteria condition, as well as their

interaction with condition (LLM vs. inclusion/exclusion criteria), will be investigated by including these variables as covariates in the linear model described above. The covariates describing the characteristics of the systematic reviews and the inclusion/exclusion criteria will also be included in the linear model in a step-wise manner to investigate their effect on starting performance. The systematic review will also be included to control for between-systematic review variation in starting performance.

The precision (or overdispersion) in starting performance across the four conditions may be explicitly modelled using a beta-binomial model (Stroup, Ptukhina, and Garai 2024). However, this may lead to convergence problems due to the limited number of unique outcome values per systematic review and condition. If so, an alternative approach will be used in which the variance in starting performance is compared across conditions using Levene’s test.

Finally, an exploratory analysis using a generalized linear mixed model (GLMM) may be performed to investigate the effect of condition on starting performance while accounting for the multilevel structure of the data (i.e., clustering by systematic review).

All analyses will be performed in R version 1.1.36 *Rsoftware*. For the exact code, see: (Ommeren 2025).

A SYNERGY metadata

B Format exported simulation results

Each simulation run is stored in a separate CSV file. Every row represents a screened paper and contains the following information:

- 1) The paper’s record ID,
- 2) The assigned label (i.e., relevant or irrelevant),
- 3) The classifier, querier, balancer and feature extractor used,
- 4) The size of the training set,
- 5) A time tag.

Furthermore, the following naming convention is used for the CSV files: `condition_run_run_IVs_n_abstracts_length_abstracts_llm_temperature.csv`. The same naming convention is used for the recall plots of each simulation run and for the generated abstracts in the LLM initialisation condition.

Finally, at the end of each run, the current values of the input parameters, the outcome variables and other relevant metadata are appended to a long format master dataframe for analysis. The columns of this dataframe are as follows:

- 1) the name of the outcome variable
- 2) the value of the outcome variable
- 3) name of the simulated dataset,
- 4) the condition,
- 5) the values of the independent variables (NaN for the control conditions):
 - (a) number of abstracts
 - (b) length of the abstracts
 - (c) temperature settings of the llm (i.e., diversity)
- 6) timestamp
- 7) the run number

This yields a data-frame containing one observation for each combination of dataset ($n=26$), condition ($n=3$), independent variables and their levels ($n=3 \times 4 \times 4 \times 5 \times 5 = 1200$), and run ($n=1$), thus with $26 \times 3 \times 1200 \times 1 = 93600$ rows, and the 12 columns enumerated above.

C Padding

It is worth noting that the simulation may stop prematurely if all the relevant records are found before the stopping rule is reached (e.g. after screening 100 records). To accurately emulate a researcher who is unaware that all the relevant records have been found and therefore continues screening until the stopping rule is reached, the simulation results were supplemented with rows containing the label 'zero' (i.e. irrelevant records) until the stopping rule would have been reached. This process is referred to as 'padding' and ensures that the final simulation results are accurate.

References

- Aamodt, Agnar and Enric Plaza (1994). "Case-based reasoning: Foundational issues, methodological variations, and system approaches". In: *AI communications* 7.1, pp. 39–59.
- Abou-Setta, Ahmed M et al. (2016). "Methods for developing evidence reviews in short periods of time: a scoping review". In: *PloS one* 11.12, e0165903.
- Achiam, Josh et al. (2023). "Gpt-4 technical report". In: *arXiv preprint arXiv:2303.08774*.
- Agrawal, Ajay, Joshua S Gans, and Avi Goldfarb (2024). "Artificial intelligence adoption and system-wide change". In: *Journal of Economics & Management Strategy* 33.2, pp. 327–337.
- Agresti, Alan (2002). *Categorical data analysis*. Hoboken.
- Alver, Sarah and Guoyi Zhang (2024). "Multiple comparisons of treatment against control under unequal variances using parametric bootstrap". In: *Journal of Applied Statistics* 51.10, pp. 1861–1877.
- Arrieta, Alejandro Barredo et al. (2020). "Explainable Artificial Intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI". In: *Information fusion* 58, pp. 82–115.
- Bastian, Hilda, Paul Glasziou, and Iain Chalmers (2010). "Seventy-five trials and eleven systematic reviews a day: how will we ever keep up?" In: *PLoS medicine* 7.9, e1000326.
- Bayer, Markus and Christian Reuter (2024). "Activellm: Large language model-based active learning for textual few-shot scenarios". In: *arXiv preprint arXiv:2405.10808*.
- Bender, Emily M et al. (2021). "On the dangers of stochastic parrots: Can language models be too big?" In: *Proceedings of the 2021 ACM conference on fairness, accountability, and transparency*, pp. 610–623.
- Bobadilla, Jesús et al. (2013). "Recommender systems survey". In: *Knowledge-based systems* 46, pp. 109–132.
- Bolukbasi, Tolga et al. (2016). "Man is to computer programmer as woman is to homemaker? debiasing word embeddings". In: *Advances in neural information processing systems* 29.
- Bommasani, Rishi (2021). "On the opportunities and risks of foundation models". In: *arXiv preprint arXiv:2108.07258*.

- Borenstein, Michael et al. (2021). *Introduction to meta-analysis*. John Wiley & Sons.
- Bornmann, Lutz and Rüdiger Mutz (2015). “Growth rates of modern science: A bibliometric analysis based on the number of publications and cited references”. In: *Journal of the association for information science and technology* 66.11, pp. 2215–2222.
- Bos, Renske (2025). “Bridging policy and technology: AI transparency in healthcare”. MA thesis.
- Brett, David and Anniek Myatt (2025). “Patience is all you need! An agentic system for performing scientific literature review”. In: *arXiv preprint arXiv:2504.08752*.
- Bretz, Frank, Torsten Hothorn, and Peter Westfall (2016). *Multiple comparisons using R*. Chapman and Hall/CRC.
- Brignardello-Petersen, Romina, Nancy Santesso, and Gordon H Guyatt (2025). “Systematic reviews of the literature: an introduction to current methods”. In: *American Journal of Epidemiology* 194.2, pp. 536–542.
- Bron, Michiel P et al. (2024). “Combining large language model classifications and active learning for improved technology-assisted review”. In: *CEUR Workshop Proceedings*. Vol. 3770. CEUR WS, pp. 77–95.
- Brown, Tom et al. (2020). “Language models are few-shot learners”. In: *Advances in neural information processing systems* 33, pp. 1877–1901.
- Bruin, Jonathan de et al. (2025). “ASReview LAB v2: Open-Source Text Screening with Multiple Agents and Oracles”. In: *Available at SSRN 5136987*.
- Brynjolfsson, Erik, Danielle Li, and Lindsey Raymond (2025). “Generative AI at work”. In: *The Quarterly Journal of Economics* 140.2, pp. 889–942.
- Burton, Andrea et al. (2006). “The design of simulation studies in medical statistics”. In: *Statistics in medicine* 25.24, pp. 4279–4292.
- Byrne, Fionn et al. (2024). “Impact of Active learning model and prior knowledge on discovery time of elusive relevant papers: a simulation study”. In: *Systematic Reviews* 13.1, p. 175.
- Caliskan, Aylin, Joanna J Bryson, and Arvind Narayanan (2017). “Semantics derived automatically from language corpora contain human-like biases”. In: *Science* 356.6334, pp. 183–186.
- Cao, Christian et al. (2025). “Automation of Systematic Reviews with Large Language Models”. In: *medRxiv*, pp. 2025–06.
- Chalmers, Iain, Larry V Hedges, and Harris Cooper (2002). “A brief history of research synthesis”. In: *Evaluation & the health professions* 25.1, pp. 12–37.
- Chen, Haichao et al. (2025). “Can large language models fully automate or partially assist paper selection in systematic reviews?” In: *British Journal of Ophthalmology*.
- Chen, Xi and Xue Zhang (2025). “Large language models streamline automated systematic review: A preliminary study”. In: *arXiv preprint arXiv:2502.15702*.
- Crevier, Daniel (1993). *AI: the tumultuous history of the search for artificial intelligence*. Basic Books, Inc.
- Cuijpers, Pim et al. (2023). “Psychological treatment of depression: A systematic overview of a ‘Meta-Analytic Research Domain’”. In: *Journal of Affective Disorders* 335, pp. 141–151.
- De Bruin, Jonathan et al. (2023). “SYNERGY-Open machine learning dataset on study selection in systematic reviews”. In: *Version V1*.
- Delgado-Chaves, Fernando M et al. (2025). “Transforming literature screening: The emerging role of large language models in systematic reviews”. In: *Proceedings of the National Academy of Sciences* 122.2, e2411962122.

- Devlin, Jacob et al. (2019). “Bert: Pre-training of deep bidirectional transformers for language understanding”. In: *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)*, pp. 4171–4186.
- Dunnett, Charles W (1964). “New tables for multiple comparisons with a control”. In: *Biometrics* 20.3, pp. 482–491.
- Europe, Horizon (2017). *Smart autopilot promises to keep flying safe — projects.research-and-innovation.ec.europa.eu*. <https://projects.research-and-innovation.ec.europa.eu/en/horizon-magazine/smart-autopilot-promises-keep-flying-safe>. [Accessed 02-02-2026].
- Eurostat (2026). *Air safety statistics in the EU — ec.europa.eu*. https://ec.europa.eu/eurostat/statistics-explained/index.php?title={A}ir_safety_statistics_in_the_{E}{U}#{C}ontext. [Accessed 01-02-2026].
- Evans, Robert (2022). “SAGE advice and political decision-making: ‘Following the science’ in times of epistemic uncertainty”. In: *Social Studies of Science* 52.1, pp. 53–78.
- Evans-Brown, Michael and Roumen Sedefov (2018). “Responding to new psychoactive substances in the European Union: early warning, risk assessment, and control measures”. In: *New psychoactive substances: Pharmacology, clinical, forensic and analytical toxicology*, pp. 3–49.
- Farrell, Henry et al. (2025). “Large AI models are cultural and social technologies”. In: *Science* 387.6739, pp. 1153–1156.
- Ferdinands, Gerbrich et al. (2023). “Performance of active learning models for screening prioritization in systematic reviews: a simulation study into the Average Time to Discover relevant records”. In: *Systematic Reviews* 12.1, p. 100.
- Fleming, Ella et al. (2025). “Position statement on artificial intelligence (AI) use in evidence synthesis across Cochrane, the Campbell Collaboration, JBI and the Collaboration for Environmental Evidence 2025”. In: *Cochrane Database of Systematic Reviews* 10.
- Funtowicz, Silvio and Jerome Ravetz (2018). “Post-normal science”. In: *Companion to environmental studies*. Routledge, pp. 443–447.
- Gartlehner, Gerald, Leila Kahwati, et al. (2025). “From promise to practice: challenges and pitfalls in the evaluation of large language models for data extraction in evidence synthesis”. In: *BMJ Evidence-Based Medicine* 30.6, pp. 385–389.
- Gartlehner, Gerald, Barbara Nussbaumer-Streit, et al. (2025). “Responsible Integration of Artificial Intelligence in Rapid Reviews: A Position Statement From the Cochrane Rapid Reviews Methods Group”. In: *Cochrane Evidence Synthesis and Methods* 3.6, e70063.
- Ge, Lixia et al. (2024). “Leveraging artificial intelligence to enhance systematic reviews in health research: advanced tools and challenges”. In: *Systematic reviews* 13.1, p. 269.
- Gouraud, Jonas, Arnaud Delorme, and Bruno Berberian (2017). “Autopilot, mind wandering, and the out of the loop performance problem”. In: *Frontiers in neuroscience* 11, p. 541.
- Grant, Maria J and Andrew Booth (2009). “A typology of reviews: an analysis of 14 review types and associated methodologies”. In: *Health information & libraries journal* 26.2, pp. 91–108.

- Haby, Michelle M et al. (2016). “What are the best methodologies for rapid reviews of the research evidence for evidence-informed decision making in health policy and practice: a rapid review”. In: *Health research policy and systems* 14.1, p. 83.
- Han, Qihua (2025). “FastSLR: An LLM-Powered Tool for Systematic Literature Review Automation”. PhD thesis. Universiteit van Amsterdam.
- Harrington, Linda (2017). “Closing the science-practice gap with technology: from evidence-based practice to practice-based evidence”. In: *AACN Advanced Critical Care* 28.1, pp. 12–15.
- Helms Andersen, T et al. (2025). “Using Artificial intelligence tools as second reviewers for data extraction in systematic reviews: a performance comparison of two AI tools against human reviewers”. In: *Cochrane Evidence Synthesis and Methods* 3.4, e70036.
- Higgins, Julian PT, Sally Green, et al. (2008). “Cochrane handbook for systematic reviews of interventions”. In.
- Hosseini, Mohammad and Serge PJM Horbach (2023). “Fighting reviewer fatigue or amplifying bias? Considerations and recommendations for use of ChatGPT and other large language models in scholarly peer review”. In: *Research integrity and peer review* 8.1, p. 4.
- Hu, Krystal (2023). *ChatGPT sets record for fastest-growing user base - analyst note*. <https://www.reuters.com/technology/chatgpt-sets-record-fastest-growing-user-base-analyst-note-2023-02-01/>. [Accessed 02-02-2026].
- Jaki, Thomas and Despina Vasileiou (2017). “Factorial versus multi-arm multi-stage designs for clinical trials with multiple treatments”. In: *Statistics in medicine* 36.4, pp. 563–580.
- Jeanquartier, Fleur et al. (2026). “Assessing the carbon footprint of language models: Towards sustainability in AI”. In: *Resources, Conservation and Recycling* 226, p. 108670.
- Jinha, Arif E (2010). “Article 50 million: an estimate of the number of scholarly articles in existence”. In: *Learned publishing* 23.3, pp. 258–263.
- Katz, Daniel Martin et al. (2024). “Gpt-4 passes the bar exam”. In: *Philosophical Transactions of the Royal Society A* 382.2270, p. 20230254.
- Khan, Muhammad Ali et al. (2025). “Collaborative large language models for automated data extraction in living systematic reviews”. In: *Journal of the American Medical Informatics Association*, ocae325.
- Khattab, Omar et al. (2023). “Dspy: Compiling declarative language model calls into self-improving pipelines”. In: *arXiv preprint arXiv:2310.03714*.
- Kim, Jin Kyu et al. (2025). “Evaluating large language models for title/abstract screening: a systematic review and meta-analysis & development of new tool”. In: *Journal of Medical Artificial Intelligence* 8, p. 34.
- Kurzer, Paulette and Darius Ornston (2023). “Follow the science: The European public health community confronts the first wave of the COVID-19 pandemic”. In: *European Policy Analysis* 9.2, pp. 101–118.
- Lai, Honghao et al. (2025). “Language models for data extraction and risk of bias assessment in complementary medicine”. In: *npj Digital Medicine* 8.1, p. 74.
- Lieberum, Judith-Lisa et al. (2025). “Large language models for conducting systematic reviews: on the rise, but not yet ready for use—a scoping review”. In: *Journal of Clinical Epidemiology*, p. 111746.
- Lika, Blerina, Kostas Kolomvatsos, and Stathes Hadjiefthymiades (2014). “Facing the cold start problem in recommender systems”. In: *Expert systems with applications* 41.4, pp. 2065–2073.

- Marcus, Gary and Ernest Davis (2019). *Rebooting AI: Building artificial intelligence we can trust*. Vintage.
- Mehand, Massinissa Si et al. (2018). “The WHO R&D Blueprint: 2018 review of emerging infectious diseases requiring urgent research and development efforts”. In: *Antiviral research* 159, p. 63.
- Mitchell, Melanie (2021). “Why AI is harder than we think”. In: *arXiv preprint arXiv:2104.12871*. — (2024). *Debates on the nature of artificial general intelligence*.
- Moosapour, Hamideh et al. (2021). “The rationale behind systematic reviews in clinical medicine: a conceptual framework”. In: *Journal of Diabetes & Metabolic Disorders* 20, pp. 919–929.
- Morris, Tim P, Ian R White, and Michael J Crowther (2019). “Using simulation studies to evaluate statistical methods”. In: *Statistics in medicine* 38.11, pp. 2074–2102.
- Mosqueira-Rey, Eduardo et al. (2023). “Human-in-the-loop machine learning: a state of the art”. In: *Artificial Intelligence Review* 56.4, pp. 3005–3054.
- Mossel, Sietse van et al. (2025). “Artificial Intelligence as a New Research Ally? Performing AI-Assisted Systematic Literature Reviews in Health Economics”. In: *PharmacoEconomics* 43.6, pp. 647–650.
- NANDA, MIT (2025). “State of AI in business 2025”. In: *Preprint at https://www.artificialintelligence-news.com/wp-content/uploads/2025/08/ai_report_2025.pdf*.
- Nick, Bostrom (2014). “Superintelligence: Paths, dangers, strategies”. In: *Strategies*.
- Noy, Shakked and Whitney Zhang (2023). “Experimental evidence on the productivity effects of generative artificial intelligence”. In: *Science* 381.6654, pp. 187–192.
- Nussbaumer-Streit, Barbara et al. (2021). “Resource use during systematic review production varies widely: a scoping review”. In: *Journal of clinical epidemiology* 139, pp. 287–296.
- O’Connor, Annette M et al. (2024). “Large language models, updates, and evaluation of automation tools for systematic reviews: a summary of significant discussions at the eighth meeting of the International Collaboration for the Automation of Systematic Reviews (ICASR)”. In: *Systematic reviews* 13.1, p. 290.
- Ollion, Étienne et al. (2024). “The dangers of using proprietary LLMs for research”. In: *Nature Machine Intelligence* 6.1, pp. 4–5.
- Ommeren, Timo Q. van (2025). *JUMP STARTING EVIDENCE SYNTHESIS*. Version 0.1.0. URL: https://github.com/timovanommeren/jump_starting_evidence_synthesis.
- Ouyang, Long et al. (2022). “Training language models to follow instructions with human feedback”. In: *Advances in neural information processing systems* 35, pp. 27730–27744.
- Page, Matthew J et al. (2021). “PRISMA 2020 explanation and elaboration: updated guidance and exemplars for reporting systematic reviews”. In: *bmj* 372.
- Panda, Deepak Kumar and Sanjog Ray (2022). “Approaches and algorithms to mitigate cold start problems in recommender systems: a systematic literature review”. In: *Journal of Intelligent Information Systems* 59.2, pp. 341–366.
- Parasuraman, Raja and Victor Riley (1997). “Humans and automation: Use, misuse, disuse, abuse”. In: *Human factors* 39.2, pp. 230–253.
- Perrigo, Billy (2023). *Exclusive: The \$2 Per Hour Workers Who Made ChatGPT Safer* — *time.com*. <https://time.com/6247678/openai-chatgpt-kenya-workers/>. [Accessed 02-02-2026].
- Powers, Mark B and Per Carlbring (2016). *Technology: bridging the gap from research to practice*.

- Radford, Alec et al. (2019). “Language models are unsupervised multitask learners”. In: *OpenAI blog* 1.8, p. 9.
- Ren, Shaolei et al. (2024). “Reconciling the contrasting narratives on the environmental impact of large language models”. In: *Scientific Reports* 14.1, p. 26310.
- Riaz, Irbaz Bin et al. (2024). “Future of Evidence Synthesis: Automated, Living, and Interactive Systematic Reviews and Meta-Analyses”. In: *Mayo Clinic Proceedings: Digital Health* 2.3, pp. 361–365.
- Sauder, Derek C and Christine E DeMars (2019). “An updated recommendation for multiple comparisons”. In: *Advances in Methods and Practices in Psychological Science* 2.1, pp. 26–44.
- Scherbakov, Dmitry et al. (2025). “The emergence of large language models as tools in literature reviews: a large language model-assisted systematic review”. In: *Journal of the American Medical Informatics Association* 32.6, pp. 1071–1086.
- Schoot, Rens van de et al. (2025). “The Hunt for the Last Relevant Paper: Blending the best of humans and AI”. In: *European Journal of Psychotraumatology* 16.1, p. 2546214.
- Shariff, Azim, Jean-François Bonnefon, and Iyad Rahwan (2017). “Psychological roadblocks to the adoption of self-driving vehicles”. In: *Nature Human Behaviour* 1.10, pp. 694–696.
- Sheng, Emily et al. (2019). “The woman worked as a babysitter: On biases in language generation”. In: *Proceedings of the 2019 conference on empirical methods in natural language processing and the 9th international joint conference on natural language processing (EMNLP-IJCNLP)*, pp. 3407–3412.
- Shyr, Cathy et al. (2025). “A statistical framework for evaluating the repeatability and reproducibility of large language models”. In: *medRxiv*, pp. 2025–08.
- Singhal, Karan et al. (2023). “Large language models encode clinical knowledge”. In: *Nature* 620.7972, pp. 172–180.
- Smela, Beata et al. (2023). “Rapid literature review: definition and methodology”. In: *Journal of market access & health policy* 11.1, p. 2241234.
- Sollini, Martina et al. (2025). “Human researchers are superior to large language models in writing a medical systematic review in a comparative multitask assessment”. In: *Scientific Reports*.
- Stevenson, Claire E et al. (2024). “Can Large Language Models generalize analogy solving like people can?” In: *arXiv preprint arXiv:2411.02348*.
- Steyvers, Mark and Aakriti Kumar (2024). “Three challenges for AI-assisted decision-making”. In: *Perspectives on Psychological Science* 19.5, pp. 722–734.
- Stroup, Walter W, Marina Ptukhina, and Julie Garai (2024). *Generalized linear mixed models: modern concepts, methods and applications*. Chapman and Hall/CRC.
- Strubell, Emma, Ananya Ganesh, and Andrew McCallum (2019). “Energy and policy considerations for deep learning in NLP”. In: *Proceedings of the 57th annual meeting of the association for computational linguistics*, pp. 3645–3650.
- Su, Xiaoyuan and Taghi M Khoshgoftaar (2009). “A survey of collaborative filtering techniques”. In: *Advances in artificial intelligence* 2009.1, p. 421425.
- Sun, Zhuanlan et al. (2024). “How good are large language models for automated data extraction from randomized trials?” In: *MedRxiv*, pp. 2024–02.
- Tan, Haocheng (2017). “A brief history and technical review of the expert system research”. In: *IOP Conference Series: Materials Science and Engineering*. Vol. 242. 1. IOP Publishing, p. 012111.

- Teijema, Jelle Jasper et al. (2025). “Large-scale simulation study of active learning models for systematic reviews”. In: *International Journal of Data Science and Analytics*, pp. 1–22.
- Thode, Lukas, Umar Iftikhar, and Daniel Mendez (2025). “Exploring the use of LLMs for the selection phase in systematic literature studies”. In: *Information and Software Technology*, p. 107757.
- Thorp, H Holden (2023). *ChatGPT is fun, but not an author*.
- Touvron, Hugo et al. (2023). “Llama: Open and efficient foundation language models”. In: *arXiv preprint arXiv:2302.13971*.
- Tran, Bao (2016). *Self-Driving Cars vs. Human Drivers: Performance, Safety, and Adoption Rates (Latest Data)* — *patentpc.com*. <https://patentpc.com/blog/self-driving-cars-vs-human-drivers-performance-safety-and-adoption-rates-latest-data>. [Accessed 01-02-2026].
- Tricco, Andrea C, Chantelle M Garritty, et al. (2020). “Rapid review methods more challenging during COVID-19: commentary with a focus on 8 knowledge synthesis steps”. In: *Journal of clinical epidemiology* 126, pp. 177–183.
- Tricco, Andrea C, Etienne V Langlois, Sharon E Straus, et al. (2017). *Rapid reviews to strengthen health policy and systems: a practical guide*. World Health Organization Geneva.
- Van De Schoot, Rens et al. (2021). “An open source machine learning framework for efficient and transparent systematic reviews”. In: *Nature machine intelligence* 3.2, pp. 125–133.
- Vaswani, Ashish et al. (2017). “Attention is all you need”. In: *Advances in neural information processing systems* 30.
- Wei, Jason et al. (2022). “Chain-of-thought prompting elicits reasoning in large language models”. In: *Advances in neural information processing systems* 35, pp. 24824–24837.
- Xia, Yu et al. (2025). “From Selection to Generation: A Survey of LLM-based Active Learning”. In: *arXiv preprint arXiv:2502.11767*.
- Xiang, Yang et al. (2025). “Exploiting Epistemic Uncertainty in Cold-Start Recommendation Systems”. In: *arXiv preprint arXiv:2502.16256*.
- Zhang, Peng and Gemma Tur (2024). “A systematic review of ChatGPT use in K-12 education”. In: *European Journal of Education* 59.2, e12599.
- Zhang, Xiangyu et al. (2023). “Cold & warm Net: addressing cold-start users in recommender systems”. In: *International Conference on Database Systems for Advanced Applications*. Springer, pp. 532–543.

Table 1: Datasets overview

Nr	Dataset	Topic(s)	Records	Included	%
1	Appenzeller-Herzog_2019	Medicine	2873	26	0.9
2	Bos_2018	Medicine	4878	10	0.2
3	Brouwer_2019	Psychology, Medicine	38114	62	0.2
4	Chou_2003	Medicine	1908	15	0.8
5	Donners_2021	Medicine	258	15	5.8
6	Hall_2012	Computer science	8793	104	1.2
7	Leenaars_2019	Psychology, Chemistry, Medicine	5812	17	0.3
8	Leenaars_2020	Medicine	7216	583	8.1
9	Meijboom_2021	Medicine	882	37	4.2
10	Menon_2022	Medicine	975	74	7.6
11	Moran_2021	Biology, Medicine	5214	111	2.1
12	Muthu_2021	Medicine	2719	336	12.4
13	Nelson_2002	Medicine	366	80	21.9
14	Oud_2018	Psychology, Medicine	952	20	2.1
15	Radjenovic_2013	Computer science	5935	48	0.8
16	Sep_2021	Psychology	271	40	14.8
17	Smid_2020	Computer science, Mathematics	2627	27	1.0
18	van_de_Schoot_2018	Psychology, Medicine	4544	38	0.8
19	van_der_Valk_2021	Medicine, Psychology	725	89	12.3
20	van_der_Waal_2022	Medicine	1970	33	1.7
21	van_Dis_2020	Psychology, Medicine	9128	72	0.8
22	Walker_2018	Biology, Medicine	48375	762	1.6
23	Wassenaar_2017	Medicine, Biology, Chemistry	7668	111	1.4
24	Wolters_2018	Medicine	4280	19	0.4

Table 2: Please note that two of the datasets included in the original SYNERGY dataset were excluded entirely due to data quality issues: Chou (2003) and Jeyaraman (2020). For one dataset (Moran, 2021), an updated version was used due to data quality issues in the original version.